AI-Based Driver Monitoring System

YOLO11s Object Detection + Transfer Learning

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Problem, Context & Solution Overview

Real-Time Driver Behavior Detection

Real-time computer vision monitors unsafe driver behaviors like phone use, smoking, missing seatbelts, and drowsiness to reduce distraction and crash risk.

Visual Design

Highlight the slide with a dark navy background and bold title. Use cyan accents on the keywords “Driver Monitoring” and “YOLO11s” to draw attention. Keep layout clean with strong contrast to reinforce a modern, AI-powered safety solution.

AI Solution

State the solution in one line: a YOLO11s-based computer vision system that analyzes live in-cabin camera input to detect unsafe driver behaviors in real time, enabling timely alerts and interventions while keeping text short and focused.

Risk Behaviors

Explain behavior categories with crisp bullets: phone use, smoking, missing seatbelt, and drowsiness detected through eye-state changes. Link each behavior to distraction, slower reactions, and higher crash risk to make the safety impact immediately clear.

Driver Safety

Current Risk

Phone Distraction

Drivers frequently handle phones for calls or messaging, pulling eyes and attention away from the road and sharply increasing collision risk.

Smoking While Driving

Lighting, holding, or dropping cigarettes forces drivers to multitask, adding hand and visual distractions that delay reaction time.

Seatbelt And Fatigue

Not wearing seatbelts worsens crash injuries, while drowsy drivers with heavy eyelids react slowly and may drift or microsleep.

VS

AI Solution

Camera Monitoring

An in vehicle or on device camera watches the driver zone, continuously capturing images or video frames without requiring manual input.

YOLO11s Detection

YOLO11s identifies phones, seatbelts, cigarettes, and driver eye states in real time, supporting both still images and live streams.

Alerts And Logging

The system triggers instant warnings and records each unsafe event, enabling rapid intervention and long term behavior analysis.

End-to-End Detection Flow & Classes

Visual detection pipeline

Each frame processed by YOLO11s: images to detections with bounding boxes, labels, confidence scores.

Detection classes & downstream use

Five key classes: open eye, closed eye, cigarette, phone, seatbelt; basis for monitoring and alerts.

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Dataset, Training Pipeline & Results

From Raw Images to Deployed Model

01

End-to-End Driver Monitoring Pipeline

Dataset to deployment: a clear, cyan-highlighted flow of six labeled, connected stages.

02

Stepwise Process Overview

Dataset, audit, YOLO11s, transfer learning, evaluation, deployment shown with minimal text and directional arrows.

Dataset Composition & Audit Findings

Metric	Count
Original images	9,884
Duplicates removed	156
Final images	9,728
Train images	5,932
Validation images	2,353
Test images	1,443
Classes	5 (Open Eye, Closed Eye, Cigarette, Phone, Seatbelt)

Dataset Volume & Class Overview

Final cleaned dataset contains 9,728 RGB JPG images across five driver-state classes with bounding boxes.

Split Summary & Data Audit Insights

Train/validation/test splits: 5,932 / 2,353 / 1,443 images; class imbalance and minor eye-label noise identified.

Training Configuration & Technical Setup

Model & Training Setup

YOLO11s with transfer learning on custom dataset, optimized for deployable driver monitoring.

Hyperparameters & Hardware

768×768 input, batch size 16, moderate epochs on Tesla T4 using pretrained weights for efficiency.

Test Performance

Global Metrics

Average Value
500

YoY Growth
10%

On 1,443 unseen test images, the model achieves 90.46% precision and 85.80% recall, with mAP@0.5 at 91.01% and mAP@0.5:0.95 at 66.99%, indicating strong detection yet room for improvement.

Quick Insights

Average Value
500

YoY Growth
10%

High precision at 90.46% means most detections are correct, while 85.80% recall shows most true objects are captured; mAP@0.5 of 91.01% reflects robust localization, and 66.99% mAP@0.5:0.95 signals potential under stricter IoU.

Class-Wise Highlights & Weaknesses

Comparison Item A

Strongest Class: Phone Usage

Phone achieves 98.25% mAP@0.5, delivering highly reliable detection of driver distraction behaviors.

VS

Comparison Item B

Weakest Class: Cigarette Detection

Cigarette reaches 45.81% mAP@0.5:0.95; small size and stricter IoU thresholds significantly reduce accuracy.

03

Demo, Failures, Future Work & Thanks

From Detection to Monitoring Output

End-to-end Monitoring Pipeline Snapshot

Deployment frame shows cyan boxes around driver, eyes, seatbelt, phone, cigarette, matching dark AI aesthetic.

From Video Stream to Actionable Safety Signals

Flow: Video → YOLO11s → Detection → Labels, confidence → Alerts, behavior logs, inputs to safety coaching systems.

Failure Cases & Limitations

Eye

Eye-related failures: open versus closed eye confusion, missed eye detections and inaccurate eye localization across varied driver poses and lighting.

Cigarette

Cigarette detection is weak under strict IoU thresholds, with small, low-contrast cigarettes often missed or poorly localized around the mouth area.

Imbalance

Many errors are driven by strong data imbalance, with far fewer rare behaviors and extreme poses compared to common, well-represented driving scenes.

Eye errors

Seatbelt

Seatbelt detection issues: duplicate seatbelt boxes on one driver, missed straps and false positives triggered by clothing folds or overlapping objects.

Labels

Annotation noise, including inconsistent or slightly shifted labels, propagates into training, making boundary cases and small objects more error-prone.

Visual

Visual complexity from lighting changes, occlusions, reflections and overlapping objects increases confusion and limits robustness of current model.

Cigarette errors

Fishbone-style overview of eye, seatbelt and cigarette detection failures, rooted in data imbalance, noisy labels and visual complexity.

Future Improvements & Closing Thanks

Future Work & Model Enhancements

Improve class balance, annotation quality, and robustness with diverse data and targeted fine-tuning.

Quote & Special Thanks

“A model detects objects. Great mentors reveal possibilities.” Deep thanks to Eng. Amr Helal and Eng. Mohamed Waleed.

Thanks

Expressing Gratitude and Appreciation in Everyday Life

Presenter： Rawan Adel